

Enterprise Digital Infrastructure – Self-assessment: Web caching

May 6, 2026

First name/Lastname _____

ID _____

A. Consider multiple HTTP requests of a browser for web objects whose size is equal to 64Kbytes. The RTT between the client and the origin web server is 4 msec and the available bandwidth is 32Mbps. Assumptions Direct interactions between the browser and the origin server. Persistent TCP connection. Secure connection.

a. The browser cache does not store any response message for the requested object. Compute the PLT.

b. The browser cache stores a response message with the following header fields:

Date: Wed 06 May 2026 07:15:00 GMT

Cache-control: private, max-age=0

Explain the header fields and compute the PLT.

c. The browser cache stores a response message with the following header fields:

Date: Wed 06 May 2026 07:15:00 GMT

Cache-control: public, max-age=7200

Explain the header fields and compute the PLT.

d. The browser cache stores a response message with the following header fields:

Date: Wed 06 May 2026 07:30:00 GMT

Expires: Fri, 8 May 2026 07:20:00 GMT

Cache-control: public

Explain the header fields and compute the PLT.

- e. The browser cache stores a response message with the following header fields:

Etag: "sgdfgna193dhqldld1100allqpe"

Cache-control: no-cache, must-revalidate

Explain the header fields and compute the PLT (specify your assumptions).

The browser cache stores a response message with the following header fields:

Date: Wed 06 May 2026 08:15:00 GMT

Last-modified: Wed 22 Apr 2026 08:15:00 GMT

Explain the header fields and compute the PLT (specify your assumptions).

- B. Consider a web page consisting of an HTML file (4Kbytes), two style sheets (6Kbytes each) and two JavaScripts (8Kbytes each). All content is stored by one origin web server. RTT between client and origin web server is 4 msec. The available bandwidth is 32Mbps.

Assumptions Direct interactions between the browser and the origin server. Persistent TCP connection. All content is cached by the browser.

Compute the minimum and maximum PLT and specify your assumptions.

- C. Consider the web page of the previous exercise. RTT between proxy and origin web server is 4 msec, available bandwidth is 32Mbps. RTT between client and proxy is 1 msec and the available bandwidth is 128Mbps.

Assumptions Access via proxy server. Persistent TCP connections. All content cached by the proxy.

Compute the minimum and maximum PLT and specify your assumptions.